

Causal Inference



Scott Cunningham

THE REMIX

Staggered DiD: Bacon + Callaway-Sant'Anna

Day 3 of 3: Morning

Scott Cunningham · Baylor University

University of Zurich · May 13, 2026

Causal Inference: The Remix · Yale University Press · forthcoming
2026

Goodman-Bacon (2021): the headline TWFE is not the ATT

The question. Did no-fault unilateral divorce laws reduce female suicide rates? Treatment timing varies across states from 1969 to 1985.

The setup. Staggered adoption + two-way fixed effects:

$$y_{ist} = \alpha_i + \gamma_t + \delta D_{ist} + \varepsilon_{ist}$$

The headline result. Stevenson-Wolfers report a meaningful negative effect: unilateral divorce reduced female suicide.

Goodman-Bacon (2021) revisited it. That headline TWFE coefficient is **not** the ATT. It's a weighted average of every 2×2 DiD you could draw with the data — including 2×2 s where one already-treated state is used as the control for a later-treated state.

Goodman-Bacon, A. (2021), "Difference-in-differences with variation in treatment timing," J. Econometrics 225(2): 254–277.

A small simulation makes the problem visible

baker.do. 1,000 firms, 4 cohorts of states, treated at 1986/1992/1998/2004. *Dynamic* treatment effects: ATT grows each year since treatment.

	The truth	TWFE estimate
Mean ATT across cohorts and post-periods	+82	−6.69

The sign flipped.

- Same data. Same DGP. Same TWFE regression you've been running.
- No econometric error in the regression itself — OLS gave you exactly what OLS gives.
- Something is wrong with the *estimand* TWFE is targeting.

Simulation: the-remix-tour/simulations/staggered/baker.do; seed 20200403.

This morning, two acts

1. **Why TWFE goes wrong (Goodman-Bacon 2021).** TWFE is a variance-weighted average of every 2×2 DiD in the data. Some of those 2×2 s use *already-treated* units as controls. When effects are dynamic, that's poison.
2. **Callaway & Sant'Anna (2021), the fix.** Forward-engineer from the target parameter $ATT(g, t)$. Estimate each one with the never-treated or not-yet-treated — never the already-treated. Then aggregate.

The afternoon-morning thread: yesterday's DR-DiD is the engine inside CS's $ATT(g, t)$.

LESSON

1. Bacon's decomposition

TWFE is a weighted average of 2×2 DiDs — and OLS picks the weights

Bacon's claim: TWFE *is* a weighted average of 2×2s

The TWFE regression with K timing groups produces an estimate $\hat{\delta}^{\text{TWFE}}$ that is *numerically identical* to a weighted average of every 2×2 DiD you can construct from the data.

The decomposition (informally)

$$\hat{\delta}^{\text{TWFE}} = \sum_{\text{every } 2 \times 2 \text{ in the data}} s_{ij} \hat{\delta}_{ij}^{2 \times 2}$$

- This is *pure* OLS algebra. No assumptions about treatment effects yet.
- It's also not surprising: any OLS coefficient is a weighted average of some underlying “comparison” — Angrist (1998), Tymon (2022) for the general result.
- The interesting questions: **which 2×2s, with what weights, and is the average we get the average we want?**

With K timing groups, there are many 2×2 s

Three timing groups $\{a, b, c\}$ plus one never-treated group U . Bacon counts the 2×2 s:

Treated vs. Never	Treated vs. Later	Treated vs. Earlier
a vs. U	a vs. b (later)	b vs. a (earlier)
b vs. U	a vs. c (later)	c vs. a (earlier)
c vs. U	b vs. c (later)	c vs. b (earlier)

- Column 1: the comparisons you intended.
- Column 2: still uses an *eventually*-treated unit as a control — it's fine, because that unit hasn't been treated yet in the comparison window.
- Column 3: uses an *already*-treated unit as a control. **This is the forbidden comparison.**

Two timing groups: k (early) and ℓ (late)

Three time windows: PRE, MID (between the two treatment dates), POST.

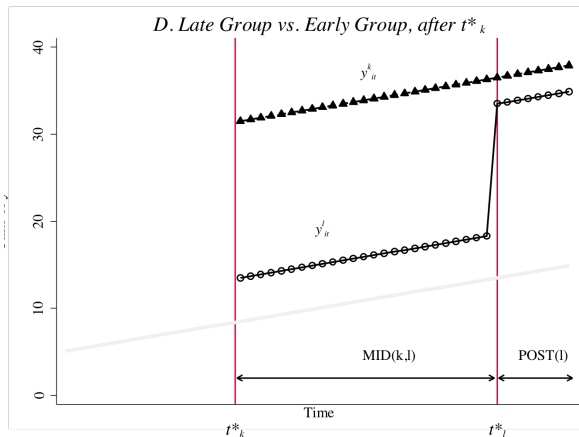
- **k vs. ℓ during MID.** Group k is post-treatment; group ℓ is still pre-treatment. **Comparison is fine** — ℓ is a not-yet-treated control.

$$\widehat{\delta}_{k\ell}^{2 \times 2, k} = (\bar{y}_k^{\text{MID}} - \bar{y}_k^{\text{PRE}}) - (\bar{y}_\ell^{\text{MID}} - \bar{y}_\ell^{\text{PRE}})$$

- **ℓ vs. k during POST.** Group ℓ is post-treatment; group k is *also* post-treatment. **The forbidden comparison** — k is now an already-treated control.

$$\widehat{\delta}_{\ell k}^{2 \times 2, \ell} = (\bar{y}_\ell^{\text{POST}} - \bar{y}_\ell^{\text{MID}}) - (\bar{y}_k^{\text{POST}} - \bar{y}_k^{\text{MID}})$$

The forbidden 2×2 , visualized



Group l treated in POST; group k already treated since MID. One side is a treated unit that should never have been a control.

The full Bacon decomposition

The setup. K timing groups, possibly one never-treated group U . TWFE estimate:

$$\widehat{\delta}^{\text{TWFE}} = \underbrace{\sum_{k \neq U} s_{kU} \widehat{\delta}_{kU}^{2 \times 2}}_{\text{treated vs. never-treated}} + \underbrace{\sum_{k \neq U} \sum_{\ell > k} s_{k\ell} \left[\mu_{k\ell} \widehat{\delta}_{k\ell}^{2 \times 2, k} + (1 - \mu_{k\ell}) \widehat{\delta}_{\ell k}^{2 \times 2, \ell} \right]}_{\text{treated vs. another-timing-group}}$$

- First sum: every timing group compared to the never-treated. Clean.
- Second sum: every *pair* of timing groups (k, ℓ) with k earlier, ℓ later, contributes two 2×2 s — one with ℓ as not-yet-treated control (fine), one with k as already-treated control (**forbidden**).
- Weights s_{kU} , $s_{k\ell}$, and the within-pair weight $\mu_{k\ell}$ come from OLS algebra, not theory.

Notation: we write the forbidden 2×2 as $\widehat{\delta}_{\ell k}^{2 \times 2, \ell}$. Bacon (2021, Eq. 10a) writes it as $\widehat{\beta}_{k\ell}^{2 \times 2, \ell}$. Same object; we follow Scott's CI-2 convention.

Where the weights come from: variance of $\tilde{D}$

After de-meaning D_{it} within unit and time, what remains is $\tilde{D}_{it}$. The Bacon weights are functions of $\widehat{\text{Var}}(\tilde{D}_{it})$ and group shares:

$$s_{kU} = \frac{n_k n_U \overline{D}_k (1 - \overline{D}_k)}{\widehat{\text{Var}}(\tilde{D}_{it})}$$

$$s_{k\ell} = \frac{n_k n_\ell (\overline{D}_k - \overline{D}_\ell) (1 - (\overline{D}_k - \overline{D}_\ell))}{\widehat{\text{Var}}(\tilde{D}_{it})}$$

$$\mu_{k\ell} = \frac{1 - \overline{D}_k}{1 - (\overline{D}_k - \overline{D}_\ell)}$$

- $\overline{D}_k$ is the share of group k 's panel spent in treatment.
- Same variance-weighting Angrist (1998) and Tymon (2022) describe generally: OLS weights treatment contrasts by treatment-status variance.

Variance is maximized in the middle of the panel

Consider the never-treated 2×2 weight, s_{kU} . The key factor: $\overline{D}_k(1 - \overline{D}_k)$.

$\overline{D}_k$ (share of panel treated)	$\overline{D}_k(1 - \overline{D}_k)$
0.1	0.09
0.3	0.21
0.5	0.25 ← max
0.7	0.21
0.9	0.09

- Groups treated in the *middle* of the panel get the highest weight.
- Same logic for $(\overline{D}_k - \overline{D}_\ell)(1 - (\overline{D}_k - \overline{D}_\ell))$.
- **Panel length affects the weights.** Trim the panel and you've changed the estimand.

So far: pure algebra. Now: potential outcomes.

What we have:

- TWFE is a weighted average of 2×2 s.
- Some of those 2×2 s use already-treated units as controls.
- OLS picks the weights based on treatment variance.

What we don't yet have:

- Whether the weighted average we get *is* the weighted ATT we want.

To answer that we need to switch from realized outcomes to potential outcomes.

The next four slides walk through a single 2×2 step by step, then substitute back into the decomposition. The result is a clean expression for the bias.

LESSON

2. The forbidden 2×2 , step by step

Add a zero, rearrange, and the bias appears

Step 1 — write down the 2×2 in observed outcomes

The 2×2 we're worried about: group ℓ (later-treated) in POST, with group k (earlier-treated) as control.

$$\hat{\delta}_{\ell k}^{2 \times 2, \ell} = [\bar{y}_{\ell}^{\text{POST}} - \bar{y}_{\ell}^{\text{MID}}] - [\bar{y}_k^{\text{POST}} - \bar{y}_k^{\text{MID}}]$$

- Both groups are *treated* in POST.
- Both groups were *treated* in MID? No — ℓ is still untreated in MID.
- So group ℓ 's “change” includes its move from untreated to treated. Group k 's “change” is from *treated* to *treated*.

Step 2 — switch to potential outcomes

Replace each observed mean with the appropriate potential-outcome mean (using the switching equation):

$$\bar{y}_{\ell}^{\text{POST}} = \mathbb{E}[Y^1 \mid \ell, \text{POST}] \quad (\text{treated})$$

$$\bar{y}_{\ell}^{\text{MID}} = \mathbb{E}[Y^0 \mid \ell, \text{MID}] \quad (\text{not yet treated})$$

$$\bar{y}_k^{\text{POST}} = \mathbb{E}[Y^1 \mid k, \text{POST}] \quad (\text{treated})$$

$$\bar{y}_k^{\text{MID}} = \mathbb{E}[Y^1 \mid k, \text{MID}] \quad (\text{already treated})$$

Notice the asymmetry. Group k is in Y^1 in *both* MID and POST. That's the problem.

Three of the four cells are Y^1 . Only one is Y^0 .

Step 3 — add zeros

We add three zero-valued terms of the form $\mathbb{E}[Y^0 \mid \cdot] - \mathbb{E}[Y^0 \mid \cdot]$:

The three zeros

1. $+\mathbb{E}[Y^0 \mid \ell, \text{POST}] - \mathbb{E}[Y^0 \mid \ell, \text{POST}]$ (add to ℓ in POST)
2. $+\mathbb{E}[Y^0 \mid k, \text{POST}] - \mathbb{E}[Y^0 \mid k, \text{POST}]$ (add to k in POST)
3. $+\mathbb{E}[Y^0 \mid k, \text{MID}] - \mathbb{E}[Y^0 \mid k, \text{MID}]$ (add to k in MID)

- Each adds zero. The 2×2 is unchanged.
- But now we can group the terms into something interpretable.

Step 4 — rearrange into ATT, parallel-trends, and ΔATT

Group the eight terms (four observed + four added zeros) into three buckets:

The forbidden 2×2 in causal pieces

$$\underbrace{\text{ATT}_\ell(\text{POST})}_{\text{what we want}} + \underbrace{\Delta Y_\ell^0(\text{POST}, \text{MID}) - \Delta Y_k^0(\text{POST}, \text{MID})}_{\text{parallel-trends bias}} - \underbrace{[\text{ATT}_k(\text{POST}) - \text{ATT}_k(\text{MID})]}_{\text{heterogeneity bias: } \Delta\text{ATT}_k} = \widehat{\delta}_{\ell k}^{2 \times 2, \ell}$$

- Bucket 1 is the ATT we want for group ℓ in the post-period.
- Bucket 2 is the standard PT bias — requires PT between ℓ and k in this window.
- Bucket 3 is **new**: it's the *change* in group k 's own ATT between MID and POST. If treatment effects grow (dynamics), this is positive — and it enters with a *minus sign*.

Step 5 — substitute back into Bacon's decomposition

The three flavors of 2×2 in causal pieces:

$$\begin{aligned}\hat{\delta}_{kU}^{2 \times 2} &= \text{ATT}_k(\text{POST}) + \Delta Y_k^0(\text{post}, \text{pre}) - \Delta Y_U^0(\text{post}, \text{pre}) \\ \hat{\delta}_{k\ell}^{2 \times 2, k} &= \text{ATT}_k(\text{MID}) + \Delta Y_k^0(\text{MID}, \text{pre}) - \Delta Y_\ell^0(\text{MID}, \text{pre}) \\ \hat{\delta}_{\ell k}^{2 \times 2, \ell} &= \text{ATT}_\ell(\text{POST}) + \Delta Y_\ell^0(\text{POST}, \text{MID}) - \Delta Y_k^0(\text{POST}, \text{MID}) \\ &\quad - \Delta \text{ATT}_k\end{aligned}$$

Plug these into Bacon's weighted sum. The TWFE probability limit:

$$\text{plim } \hat{\delta}^{\text{TWFE}} = \text{VWATT} + \text{VWPT} - \Delta \text{ATT}$$

Bacon (2021, Eq. 15) writes VWCT (variance-weighted common trends). Same object as our VWPT.

What the decomposition tells us

$$\text{plim } \hat{\delta}^{\text{TWFE}} = \text{VWATT} + \text{VWPT} - \Delta\text{ATT}$$

- **VWATT** — a variance-weighted ATT. Not the ATT *you* want unless you specifically wanted variance-weighted averaging. The weights depend on *panel length* (a researcher choice).
- **VWPT** — variance-weighted parallel-trends bias. Zero only if PT holds across *every* pairwise comparison, including between treated groups.
- **ΔATT** — the dynamics term. If ATT grows with time-since-treatment, this is *positive* and enters with a *minus sign* — attenuating TWFE toward zero, or even **flipping the sign**.

That's exactly what happens in baker.do: dynamics big enough that $-\Delta\text{ATT}$ swamps the rest, flipping +82 to -6.69.

LESSON

3. baker.do, in code

Set the seed, write the DGP, compute the truth, run TWFE, watch it fail

The panel: 30 explicit lines, because repetition *is* the lesson

```
set obs 40
gen state = _n
expand 25
bysort state: gen firms = runiform(0,5)
expand 30
sort state
bysort state firms: gen year = _n

replace year = 1980 if year==1
replace year = 1981 if year==2
replace year = 1982 if year==3
replace year = 1983 if year==4
replace year = 1984 if year==5
replace year = 1985 if year==6
replace year = 1986 if year==7 // cohort 1 first treated here
replace year = 1987 if year==8
replace year = 1988 if year==9
... (22 more explicit lines, one per year, through 2009) ...
```

A loop would be shorter. The 30 lines are the lesson: each is one substitution, one beat. See [simulations/README.md §1](#).

The DGP, in 12 more explicit lines

```
set seed 20200403

* Group-specific TE means (heterogeneous across cohorts)
gen te1 = rnormal(10, 0.04) // cohort 1986
gen te2 = rnormal( 8, 0.04) // cohort 1992
gen te3 = rnormal( 6, 0.04) // cohort 1998
gen te4 = rnormal( 4, 0.04) // cohort 2004
gen te = cond(group==1, te1, cond(group==2, te2, cond(group==3, te3, te4)))

* Y(0) and Y(1) under DYNAMIC effects (te grows since treatment)
gen y0 = firms + n + e
gen te_dynamic = te*treat*(year - treat_date + 1)
gen y = y0 + te_dynamic
```

One step per line, named variables, no loops. Full file: simulations/staggered/baker.do.

The truth, computed before running anything

```
* The truth (averaged across cohorts and post-periods)  
su te_dynamic if treat == 1  
display "True average ATT (dynamic) = " r(mean)
```

Output: True average ATT (dynamic) = 82

Why 82? The dynamic effect for group 1 averages $(10 + 20 + \dots + 240)/24$. Combined across cohorts and post-periods, the simple average is 82.

We computed the truth from the DGP. Now compare it to what TWFE produces.

TWFE, applied

```
areg y i.year treat, a(id) robust
```

	Truth	TWFE
$\widehat{ATT}$	+82	−6.69***

Sign-flipped. Extreme dynamics — the $-\Delta ATT$ term swamps VWATT.

The Bacon decomposition of that -6.69

DD comparison	Weight	Avg DD estimate
Earlier-treated vs. later-treated-(as-control)	0.500	+51.80
Later-treated vs. earlier-treated-(as-control)	0.500	-65.18

$$(0.5)(+51.80) + (0.5)(-65.18) = -6.69.$$

Read it. Half the weight is on a 2×2 averaging -65 — a 2×2 that uses an already-treated group as control. **That's the forbidden comparison doing the damage.**

Bacon's own decomposition of Stevenson-Wolfers

We opened with Stevenson-Wolfers. Bacon ran the same decomposition on their actual data.

$$\hat{\beta}_{SW}^{DD} = -3.08 \quad (156 \text{ components})$$

- The late-vs-early-treated 2×2 s average **+3.51** — the wrong sign, pulled by treatment-effect heterogeneity over time.
- Drop those terms: the rest averages -5.44 , close to the event-study average (-4.92).
- The single TWFE coefficient of -3.08 is masking a $+3.51$ vs. -5.44 split. **Same disease as baker.do, less extreme.**

Goodman-Bacon (2021, §4 and Fig. 5). The SW headline survives qualitatively but its magnitude is contaminated.

LESSON

4. Visualizing staggered structure

What the cells look like, what TWFE averages over

Constant-TE worksheet

Year	ATT(1986,t)	ATT(1992,t)	ATT(1998,t)	ATT(2004,t)
1980–1985	0	0	0	0
1986–1991	10	0	0	0
1992–1997	10	8	0	0
1998–2003	10	8	6	0
2004–2009	10	8	6	4
ATT(<i>g</i>)	10	8	6	4

- **Simple ATT** (uniform avg over colored cells) = 8.0.
- **VWATT** (TWFE's target) $\neq$ 8.0 — OLS weights down edge cohorts via $\bar{D}(1 - \bar{D})$.

Even with constant effects per cohort, TWFE doesn't give the simple ATT.

Dynamic-TE worksheet

Year	ATT(1986,t)	ATT(1992,t)	ATT(1998,t)	ATT(2004,t)
1986	10	0	0	0
1992	70	8	0	0
1998	130	56	6	0
2004	190	104	42	4
2009	240	144	72	24

ATT grows each year since treatment (the dynamic DGP). Simple ATT (uniform avg over colored cells) ≈ 82 ; TWFE = -6.69 .

The $-\Delta$ ATT bias is green doing the damage. Late cohorts use the already-treated earlier cohorts as “controls” — whose outcomes are growing.

LESSON

5. Callaway & Sant'Anna

Forward-engineer from $ATT(g, t)$, never use the already-treated as control

The forward-engineering principle

Bacon told us what reverse-engineering from TWFE buys you: a variance-weighted average of every 2×2 in the data, including the forbidden ones.

The Callaway-Sant'Anna (2021) inversion:

1. **Start** with the target parameter you actually want: $ATT(g, t)$ — the average treatment effect on cohort g in calendar period t .
2. **State** the identification assumptions formally.
3. **Derive** an estimator directly from those assumptions.
4. **Aggregate** the building blocks into whatever summary you want.

Each $ATT(g, t)$ is its own clean 2×2 problem. Yesterday afternoon's DR-DiD is exactly the engine that estimates one.

ATT(g, t) — the building block

Definition

$$\text{ATT}(g, t) = \mathbb{E}[Y_{i,t}(1) - Y_{i,t}(0) \mid G_i = g]$$

- $G_i = g$ = “unit i was first treated in period g .” (Cohort identifier.)
- One $\text{ATT}(g, t)$ for every (cohort, calendar-period) pair where the cohort is treated.
- In baker.do: 4 cohorts $\times$ (varying numbers of post-periods) = 60 $\text{ATT}(g, t)$ cells in the grid.
- Each cell is a single 2×2 estimand. **The aggregation is a separate, downstream choice.**

Three flavors of parallel trends for staggered settings

The 2×2 PT assumption generalizes three ways:

Assumption	Comparison group	When to use
PT-GT-Nev	Never-treated only	Conservative; needs a never-treated to exist
PT-GT-NYT	Not-yet-treated	Authors' preferred — middle ground
PT-GT-all	All groups, all periods	Strongest assumption; imposes parallel pre-trends

- NYT lets late-treated groups serve as controls for earlier-treated groups in their pre-treatment window. Never as already-treated controls.
- The forbidden 2×2 of Bacon is *ruled out by construction*.

Baker, Callaway, Cunningham, Goodman-Bacon, Sant'Anna (2025), "DiD: A Practitioner's Guide."

The estimator — and yesterday's DR is the recommended one

Three options for the building-block estimator of each $ATT(g, t)$:

1. **OR.** Heckman-Ichimura-Todd: model $\Delta Y(0)$ on X , impute the treated counterfactual.
2. **IPW.** Abadie (2005): weight controls by $p(X)/(1 - p(X))$.
3. **DR.** Sant'Anna-Zhao (2020). Consistent if *either* model is correct.

The practitioner's-guide recommendation: “Favor doubly robust.”

$$\hat{\delta}_{DR}(g, t) = \mathbb{E} \left[w_{DR}(X) (\Delta Y - \hat{\mu}_0(X)) \right]$$

The DR-DiD you proved yesterday afternoon. Same engine, applied cell-by-cell to every feasible (g, t) .

Aggregation is a downstream choice, not an estimation choice

Once you have $\{\widehat{ATT}(g, t)\}$, build whatever summary the policy question demands:

- **Simple ATT.** Uniform average over all post-treatment (g, t) cells.
- **Group ATT, $ATT(g)$.** Average each cohort's own post-treatment cells. "What did the policy do to cohort g on average?"
- **Event-time ATT, $ATT_{es}(e)$.** Average across cohorts at fixed event time e since treatment. "What's the dynamic profile?"
- **Calendar-time ATT.** Average across cohorts within a calendar period. "What happened in 2002, on average?"

Inference. Bootstrap, with uniform confidence bands when reporting an event-study (the `did` R package does this by default).

baker_cs.do — the implementation

```
* Get the same baker data, fit CS with DR-DiD
ssc install csdid, replace
ssc install drdid, replace

csdid y, ivar(id) time(year) gvar(treat_date) ipw long2

* Four aggregations, four answers:
csdid_estat simple // simple ATT across all cells
csdid_estat group // ATT(g) per cohort
csdid_estat calendar // ATT by calendar period
csdid_estat event // event-time profile
```

Building blocks first, aggregation second.

collegio-carlo-alberto/labs/Baker/Solutions/baker_cs.do.

The headline: CS recovers the truth

	Truth	TWFE	CS (DR-DiD)
Feasible ATT (pre-2004 sample)	68.33	26.81***	68.34***

- The “feasible” sample drops the 2004 cohort so a never-treated comparison exists.
- TWFE is no longer sign-flipped (no universally-treated tail), but it still attenuates — still has VWATT weights, still uses early-treated cohorts as controls.
- CS lands on the truth to two decimals.

Not every $ATT(g, t)$ is feasible

CS can only estimate $ATT(g, t)$ if there is a valid comparison group for (g, t) :

- Under **PT-GT-Nev**: need a never-treated unit observed at t .
- Under **PT-GT-NYT**: need a not-yet-treated unit observed at t .

In baker.do: The last cohort is treated in 2004. After 2004, every unit is treated. **No comparison units exist for $ATT(g, t > 2004)$.**

Two responses:

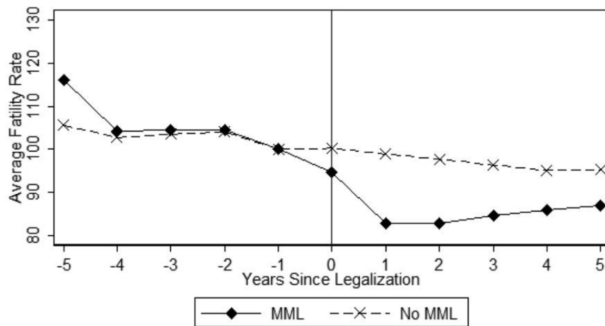
- Restrict to the feasible sample. CS will tell you which (g, t) it dropped.
- Or, if you really want estimates for the universally-treated tail, use a different framework (Imputation: Borusyak et al. 2024; Wooldridge 2021) that doesn't require a comparison group.

LESSON

6. Sun & Abraham (2021)

Bacon was static. Event-studies have their own contamination.

Researchers love event studies — and TWFE will lie to you on them



Anderson et al. (2013): raw traffic-fatality rates for treated states (re-centered to treatment date) vs. controls. The classic dynamic-effects display.

The classic dynamic-TWFE event-study spec

Researchers want to plot pre-treatment placebo coefficients and post-treatment dynamic effects. The standard regression:

$$Y_{i,t} = \alpha_i + \delta_t + \sum_{g \in G} \mu_g \mathbf{1}\{t - E_i \in g\} + \varepsilon_{i,t}$$

- E_i is unit i 's treatment date. $t - E_i$ is event time (relative-to-treatment).
- G is the set of event-time bins (leads and lags) included.
- Coefficient μ_g is supposed to be the average treatment effect at event-time bin g .

Bacon told us static TWFE is biased. Sun & Abraham (2021) ask: *what about the dynamic version?*

SA's proposition: μ_g is contaminated by other leads and lags

Even under parallel trends and no anticipation, the TWFE event-study coefficient on bin g is:

$$\begin{aligned}\mu_g = & \underbrace{\sum_{l \in g} \sum_e w_{e,l}^g \text{CATT}_{e,l}}_{\text{the target (own bin)}} \\ & + \underbrace{\sum_{g' \neq g} \sum_{l' \in g'} \sum_e w_{e,l'}^g \text{CATT}_{e,l'}}_{\text{contamination from other included bins}} \\ & + \underbrace{\sum_{l' \in g^{\text{excl}}} \sum_e w_{e,l'}^g \text{CATT}_{e,l'}}_{\text{contamination from excluded bins}}\end{aligned}$$

Other-bin weights sum to zero — but only cancel under treatment-effect homogeneity. With heterogeneous CATTs, the contamination doesn't cancel.

The disturbing corollary: pre-period μ_g can be nonzero even when PT holds

Under PT + no anticipation but *not* homogeneity:

- A pre-period lead μ_{-2} should be zero if pre-trends are truly parallel.
- In TWFE it isn't — it picks up CATTs from *post-treatment* bins through the cross-bin weights.
- So a “failed placebo” may just be the cross-bin contamination.
- And a “passing placebo” may just be cancellation, not evidence of PT.

Lesson: Don't drop excluded bins arbitrarily. Lagged effects contaminate through the excluded bins.

Same disease as Bacon's static decomposition, but applied to leads and lags — so it affects the entire plot you're showing the audience.

SA's fix: the interaction-weighted estimator (3 steps)

1. **Estimate each $\widehat{\delta}_{e,l}$ separately.** A saturated regression with cohort $\times$ event-time interactions, using the never-treated (or last-treated) as control:

$$Y_{i,t} = \alpha_i + \lambda_t + \sum_{e \notin C} \sum_{l \neq -1} \delta_{e,l} [\mathbf{1}\{E_i = e\} \cdot D_{i,t}^l] + \varepsilon_{i,t}$$

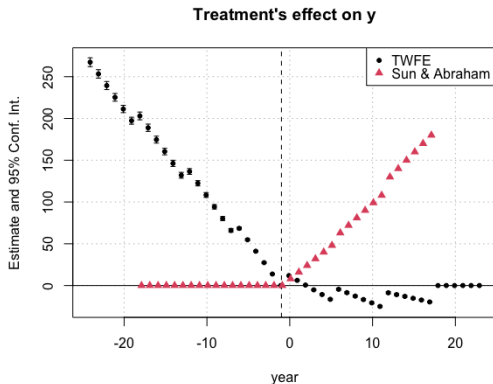
2. **Estimate cohort shares** in the relevant periods: $\widehat{\Pr}(E_i = e \mid E_i \in [-l, T - l])$.
3. **Aggregate** the $\widehat{\delta}_{e,l}$ using the cohort shares as weights.

Same logic as CS. Estimate cohort-event-time cells first; aggregate after.

Forward-engineer the parameter. *Software: `eventstudyinteract` (Stata, SSC);*

`fixest::sunab` (R).

TWFE event study vs. SA, side by side



TWFE leads slope downward pre-treatment — the cross-bin contamination. SA's pre-period sits at zero; lags trace the dynamic profile.

LESSON

7. de Chaisemartin & D'Haultfœuille

Negative weights on *unit-level* effects, plus the on-off case

dCdH (2020) adds three things to the staggered toolkit

1. **A different decomposition.** Bacon decomposed TWFE into 2×2 building blocks with *positive* weights. dCdH decomposes TWFE into *unit-level* treatment effects — where weights can be **negative**.
2. **The on-off case.** CS and SA assume irreversible treatment. dCdH handles units that switch into and out of treatment (think: a state policy that was repealed). Roe v. Wade adopted in 1973, overturned in 2022.
3. **A diagnostic.** Run the TWFE you were going to run, then count the *share of unit-period treatment effects* that receive negative weights. Report the sign-flip threshold.

The dCdH decomposition: weights on unit-level effects

Let $\Delta_{i,t}^g = Y_{i,t}^1 - Y_{i,t}^\infty$ be the *unit-level* treatment effect for unit i in period t . Under PT, no anticipation, and SUTVA:

Theorem (dCdH 2020)

$$\delta^{\text{TWFE}} = \mathbb{E} \left[\sum_{i,t: D_{i,t}=1} \frac{1}{N^T} w_{i,t} \Delta_{i,t}^g \right], \quad \sum_{i,t: D_{i,t}=1} \frac{w_{i,t}}{N^T} = 1, \quad w_{i,t} \text{ can be negative.}$$

- Weights come from the OLS residual of $D_{i,t}$ on unit and time FE.
- Bacon's weights are on 2×2 s (and always positive). dCdH's weights are on *individual treatment effects* (and may be negative).

Negative weights break the “no sign-flip property”

Thought experiment. Suppose every unit gains from the treatment ($\Delta_{i,t}^g > 0$ for all i, t).

- If all $w_{i,t} > 0$: $\hat{\delta}^{\text{TWFE}}$ is a positive weighted average of positive effects. Sign matches.
- If some $w_{i,t} < 0$: nothing prevents $\hat{\delta}^{\text{TWFE}}$ from being **negative** — even though every unit gained.

This is what we saw in baker.do. Truth +82, TWFE −6.69. Some unit-level effects got large negative weights; they swamped the positive-weight contributions. *Pre-dCdH, “OLS is a weighted average” was reassuring. dCdH made it disturbing.*

When the negative weights bite (and when they don't)

The diagnostic dCdH recommends running, before you publish anything:

1. Fraction of unit-period treatment effects receiving *negative* weight in your TWFE.
2. The minimum amount of heterogeneity in $\Delta_{i,t}^g$ that would flip the sign of $\hat{\delta}^{\text{TWFE}}$.

Rules of thumb:

- Large never-treated group $\Rightarrow$ fewer negative weights, smaller magnitudes.
- Treatment dates packed close together $\Rightarrow$ also fewer.
- Long panel with a few early-treated cohorts $\Rightarrow$ the worst case.

Software: `twowayfeweights` (Stata, SSC; R via the `TwoWayFEWeights` package) reports both diagnostics on your specific dataset.

The on-off case: `did_multipligt` for switching treatment

Suppose treatment can switch on *and* off (Roe v. Wade 1973 → Dobbs 2022; minimum-wage rollbacks; tax cuts that sunset).

- CS and SA both assume **irreversible** treatment — they cannot estimate effects in this setting.
- dCdH propose a different estimator built on *instantaneous* effects between adjacent periods, comparing units that just switched against units whose treatment status is unchanged.
- Equivalent to CS & SA when treatment is irreversible (with their particular weighting), but *also* handles the on-off case.

Software: `did_multipligt` (Stata); the `DIDmultipligt` R package.

The unifying lesson across Bacon, SA, and dCdH

What makes weights “good”

The economic question is: *what parameter do you want?*

OLS doesn't ask. OLS picks weights from the residual variance of D .

- Bacon: TWFE = weighted average of 2×2 s. Weights positive but variance-driven.
- SA: event-study μ_g = weighted average of $\text{CATT}_{e,l}$ across bins. Cross-bin contamination.
- dCdH: TWFE = weighted average of unit-level $\Delta_{i,t}^g$. **Weights can be negative.**

All three say the same thing: If you let OLS pick the weights, you've let it pick your economic question for you. Forward-engineer instead.

LESSON

8. Borusyak-Jaravel-Spiess imputation

The Heckman-Ichimura-Todd move, lifted to fixed effects

All causal inference is imputation

Imbens & Rubin (2015), *Causal Inference*

“At some level, all methods for causal inference can be viewed as imputation methods, although some more explicitly than others.”

What's the missing thing? Always Y^0 for the treated. What we never observe.

What's the estimator? A model for Y^0 fit on the people who reveal it (the untreated), then plugged in for the treated.

You already saw one today. Yesterday afternoon's OR-DiD was imputation. BJS is the same trick, with a different model.

Yesterday's HIT / OR-DiD recipe (recap)

Heckman, Ichimura, Todd (1997). The conditional-PT version of imputation:

1. **Fit the model** of the untreated trend on X , using only $D = 0$:

$$\hat{\mu}_0(X) = \widehat{\mathbb{E}}[\Delta Y \mid D=0, X]$$

2. **Impute** each treated unit's counterfactual trend $\hat{\mu}_0(X_i)$.

3. **Average** the residual on the treated:

$$\hat{\delta}_{\text{OR}} = \frac{1}{N_1} \sum_{i:D_i=1} (\Delta Y_i - \hat{\mu}_0(X_i))$$

The counterfactual model is on ΔY , indexed by X .

The BJS (2024) recipe, side-by-side

Borusyak, Jaravel, Spiess (2024). Same imputation logic, but the counterfactual model is on *levels* $Y(0)$ indexed by *fixed effects*:

1. **Fit the model** of untreated $Y(0)$ on unit FE + time FE (and any pre-treatment X), using only $D = 0$ observations:

$$Y(0)_{it} = \alpha_i + \lambda_t + X'_{it}\delta + \varepsilon_{it}$$

2. **Impute** the counterfactual for each treated (i, t) : $\hat{Y}_{it}^0 = \hat{\alpha}_i + \hat{\lambda}_t + X'_{it}\hat{\delta}$.
3. **Average** the residual on the treated unit-periods:

$$\hat{\tau}_{it} = Y_{it}^1 - \hat{Y}_{it}^0, \quad \hat{\delta}_W = \sum_{(i,t) \in \Omega_1} w_{it} \hat{\tau}_{it}$$

Same three moves as HIT. Only the counterfactual model differs.

Why BJS avoids the forbidden contrast

The fixed-effects regression in Step 1 is fit on **untreated observations only**.

- No treated observation enters the model that predicts $Y(0)$.
- Already-treated units contribute their pre-treatment periods only (those are still untreated).
- The fixed effects propagate across time, so $\hat{\alpha}_i$ for unit i can still be estimated even if i is eventually treated — using i 's own pre-treatment observations.

Compare to TWFE. TWFE fits δ *and* the fixed effects on *all* observations simultaneously — so a treated unit's post-period influences the estimated FE, which feeds back into the residual that defines $\hat{\delta}$. That's the forbidden-contrast channel.

BJS just refuses to let that happen. Same model class, different fitting sample.

BJS compared with TWFE, CS, SA on the same DGP

Table 3: Efficiency and Bias of Alternative Estimators

Horizon	Estimator	Baseline simulation		More pre-periods	Heterosk. residuals	AR(1) residuals	Anticipation effects
		Variance (1)	Coverage (2)	Variance (3)	Variance (4)	Variance (5)	Bias (6)
$h = 0$	Imputation	0.0099	0.942	0.0080	0.0347	0.0072	-0.0569
	DCDH	0.0140	0.938	0.0140	0.0526	0.0070	-0.0915
	SA	0.0115	0.938	0.0115	0.0404	0.0066	-0.0753
$h = 1$	Imputation	0.0145	0.936	0.0111	0.0532	0.0143	-0.0719
	DCDH	0.0185	0.948	0.0185	0.0703	0.0151	-0.0972
	SA	0.0177	0.948	0.0177	0.0643	0.0165	-0.0812
$h = 2$	Imputation	0.0222	0.956	0.0161	0.0813	0.0240	-0.0886
	DCDH	0.0262	0.958	0.0262	0.0952	0.0257	-0.1020
	SA	0.0317	0.950	0.0317	0.1108	0.0341	-0.0850
$h = 3$	Imputation	0.0366	0.928	0.0255	0.1379	0.0394	-0.1101
	DCDH	0.0422	0.930	0.0422	0.1488	0.0446	-0.1087
	SA	0.0479	0.952	0.0479	0.1659	0.0543	-0.0932
$h = 4$	Imputation	0.0800	0.942	0.0546	0.3197	0.0773	-0.1487
	DCDH	0.0932	0.950	0.0932	0.3263	0.0903	-0.1265
	SA	0.0932	0.954	0.0932	0.3263	0.0903	-0.1265

Notes: See Section 4.6 for a detailed description of the data-generating processes and reported statistics.

TWFE event-study: biased pre-period leads, attenuated lags. CS, SA, BJS all recover the dynamic profile. Three different counterfactual models, same answer.

When BJS, when CS, when SA, when dCdH?

- **CS (DR).** You want $ATT(g, t)$ building blocks. You have covariates that matter for conditional PT. **First choice when standard staggered.**
- **SA.** You specifically want an event-study plot. Equivalent to CS with never-treated; gives closed-form SEs.
- **BJS.** You want a single $\hat{\tau}_{it}$ for every treated unit-period — maximum flexibility, can compute any aggregate. Comfortable assuming the parametric $Y(0)$ model.
- **dCdH.** Your treatment switches *on and off*. CS, SA, BJS all assume irreversibility; dCdH does not.

Practitioner's guide framing: all four are forward-engineered estimators. Pick the one that matches the parameter you want and the assumptions you can defend.

LESSON

9. Synthesis

What's the same, what's different, what to do tomorrow morning

Yesterday afternoon plus this morning = one workflow

- **Yesterday (covariates).** Conditional PT $\Rightarrow$ DR-DiD. Two chances to be right (OR model or pscore model).
- **This morning (staggered).** $ATT(g, t)$ building blocks, estimated cell-by-cell, then aggregated.
- **The combination.** Each $ATT(g, t)$ is a 2×2 problem. The recommended way to estimate it is DR-DiD with whatever X you need for conditional PT. Then aggregate.

One sentence

Forward-engineer the target, estimate the cells with DR, aggregate with non-negative weights.

Basically never.

- TWFE = ATT only when (a) treatment effects are homogeneous across cohorts, AND (b) treatment effects don't change over time within a cohort.
- Both fail in almost every applied setting.
- Even when they hold, the variance weights are still strange (depend on panel length).
- **If you must report it, also report CS, SA, BJS, or dCdH — and the dCdH negative-weight diagnostic. Explain the differences.**

Baker, Callaway, Cunningham, Goodman-Bacon, Sant'Anna (2025): "Do not use TWFE."

A student asks: “But what about my paper from last year that used TWFE?”

A reasonable thing to do, in order:

1. **Run the Bacon decomposition** (`ddtiming`; `bacondecomp` in R). See whether the forbidden 2×2 s carry weight.
2. **Run the dCdH diagnostic** (`twowayfeweights`). Sizeable negative-weight share $\Rightarrow$ headline is suspect.
3. **Re-estimate with CS, SA, or BJS**. Compare the new headline to TWFE.
4. **If they disagree:** the difference *is* the $-\Delta\text{ATT} / \text{VWPT}$ / negative-weight contamination. Trust the design-based estimator.
5. **If they agree:** in your setting, dynamics and heterogeneity didn't bite. Report both, explain.

Goodman-Bacon, Sant'Anna, Whitten (2026) gives a refined re-analysis framework for pre-trends-parallel settings.

Lab: rerun baker, rerun baker_cs, compare

- **Step 1.** Run `simulations/staggered/baker.do` to (re)generate the dynamic-TE dataset. Confirm the truth is +82 and TWFE is -6.69 .
- **Step 2.** Run the Bacon decomposition (`ddtiming`). Verify the two weights and the two 2×2 averages reproduce the textbook table.
- **Step 3.** Run `baker_cs.do`: CS with DR-DiD, plus four aggregations (`simple`, `group`, `calendar`, `event`).
- **Step 4.** Drop the 2004 cohort. Re-run TWFE: notice it stops sign-flipping (no longer have an universally-treated tail). Re-run CS: now lands on the truth.
- **Step 5.** Write a one-paragraph summary: which assumption is doing the work, and what would have to be true for TWFE to be correct?

Walkthrough: `zurich/labs/Baker/README.md`. Code: `simulations/staggered/baker.do`, plus the CS files Scott will provide.

Tomorrow afternoon

- **Honest event-studies.** Roth (2022) on conditioning on pretest passes. Rambachan & Roth (2023) partial-identification bounds.
- **Continuous and multi-valued treatment.** Dose-specific ATT, average causal response (Callaway, Goodman-Bacon, Sant'Anna 2024).
- **Goodman-Bacon, Sant'Anna, Whitten (2026)** — a refined re-analysis framework for pre-trends-parallel settings.
- **Two-stage DiD** (Gardner 2021) — the GMM cousin of BJS, with the same forward-engineered architecture.

Reading for tonight: *Baker, Callaway, Cunningham, Goodman-Bacon, Sant'Anna (2025)*
“Difference-in-Differences Designs: A Practitioner’s Guide.”

Forward-engineer.

*Start with what you want to estimate.
Match the estimator to the assumption.
Aggregate at the end, not the start.*

Scott Cunningham · University of Zurich · May 13, 2026